EXPERIMENT 38: DETERMINATION OF ACUTE TOXICITY (LD50)

40 STANDARD IN-COURSE MCQs

1. The primary objective of Experiment 38 is to determine:

A. The chronic carcinogenic potential of a compound B. The median lethal dose (LD_50) of a test substance in mice C. The minimum inhibitory concentration against bacteria D. The rate of renal tubular secretion

Answer: B

2. What is the standard pharmacological definition of LD_50?

A. The lowest dose of a drug that produces any observable toxic symptom B. The statistically derived single dose of a substance that kills 50% of an animal population under defined conditions C. The dose of a drug that cures 50% of infected animals D. The maximum tolerated dose with zero animal deaths

Answer: B

3. What experimental animals are designated in the protocol on Page 173?

A. Adult guinea pigs B. Albino mice (of approximately the same body weight) C. Adult rabbits D. Wistar rats

Answer: B

4. What pre-experimental fasting requirement is specified for the mice on Page 173?

A. Fasted 2 hours with no water B. Fasted 12 hours with water ad libitum C. Fasted 48 hours without food or water D. No fasting required

Answer: B

5. Why is water allowed "ad libitum" while food is withheld for 12 hours?

A. To cause severe dehydration B. To empty the digestive tract without producing hypovolemic dehydration or metabolic collapse C. To accelerate renal failure D. To induce respiratory depression

Answer: B

6. Why must the experimental animals in all dosing groups be of approximately the same weight?

A. Syringes only work for one specific weight B. Weight differences alter surface-to-volume ratio, metabolic rate, and drug volume, causing artifactual variation C. Heavy animals never absorb intraperitoneal drugs D. Light animals are immune to toxicity

Answer: B

7. What pattern of dose progression is typically used when setting up acute toxicity groups?

A. Arithmetic progression (e.g., 10, 11, 12, 13 mg/kg) B. Geometric (logarithmic) progression (e.g., 20, 40, 80, 160, 320 mg/kg) C. Random arbitrary doses D. Administering the same dose to all groups

Answer: B

8. What initial observation time window is specified to count acute deaths on Page 173?

A. 5 minutes B. Within 2 hours after injection C. Exactly 1 week D. 30 days

Answer: B

9. What extended observation period is standard in acute toxicity testing to detect delayed mortality?

A. 4 hours B. 24 hours (up to 7–14 days) C. 1 month D. 1 year

Answer: B

10. In addition to counting deaths, what other parameters must be recorded?

A. Color of the animal cage B. General behavioral changes, autonomic signs, motor alterations, convulsions, and respiratory distress C. Room air pressure D. Amount of bedding material soiled

Answer: B

11. Which behavioral sign in mice indicates central spinal and sympathetic stimulation often seen with morphine or convulsants?

A. Loss of whiskers B. Straub tail phenomenon (rigid dorsiflexion of the tail) C. Sleeping in the corner D. Excessive drinking of water

Answer: B

12. In the graphical method of Miller and Tainter for LD_50 determination, percentage mortality is converted into:

A. Decibels B. Probit units (probability units) C. Natural logarithms of weight D. Linear millimeters

Answer: B

13. On a probit versus log-dose plot, what probit value corresponds exactly to 50% mortality?

A. Probit 3.0 B. Probit 4.0 C. Probit 5.0 D. Probit 6.0

Answer: C

14. What probit value corresponds to 16% mortality (mean minus one standard deviation)?

A. Probit 3.0 B. Probit 4.0 C. Probit 5.0 D. Probit 6.0

Answer: B

15. What probit value corresponds to 84% mortality (mean plus one standard deviation)?

A. Probit 4.0 B. Probit 5.0 C. Probit 6.0 D. Probit 7.0

Answer: C

16. Why are percentages converted to probits in quantal dose-response analysis?

A. To make the numbers larger B. To transform the sigmoidal cumulative frequency curve into a straight line for accurate linear regression C. To eliminate experimental errors D. To convert grams into milligrams

Answer: B

17. In the arithmetic method of Karber (Behrens-Karber), the formula for calculating LD_50 is:

A. LD_50 = LD_100 - [sum(z * d) / n] B. LD_50 = LD_0 + [sum(z * d) * n] C. LD_50 = (LD_100 + LD_0) / 2 D. LD_50 = sum(z * d) / n

Answer: A

18. In Karber's formula, what does "z" represent?

A. Total number of animals tested B. The mean number of dead animals between two consecutive dose levels C. The difference between body weights D. The time of death in minutes

Answer: B

19. In Karber's formula, what does "d" represent?

A. Duration of the experiment B. The difference between two consecutive dose levels C. The number of survivors D. The diameter of the syringe needle

Answer: B

20. In Karber's formula, what does "n" represent?

A. Number of animals per group B. Total number of dead animals C. Number of groups tested D. Concentration of the drug solution

Answer: A

21. In Lorke's method (1983) for acute toxicity testing, how many phases are conducted?

A. One phase only B. Two phases (Phase 1 screening followed by Phase 2 specific dosing) C. Five phases D. Ten phases

Answer: B

22. In Lorke's Phase 1, what three initial doses are typically administered to groups of 3 mice each?

A. 1, 2, 3 mg/kg B. 10, 100, and 1,000 mg/kg C. 50, 60, 70 mg/kg D. 2,000, 4,000, 6,000 mg/kg

Answer: B

23. How is the final LD_50 calculated using Lorke's formula?

A. Arithmetic mean of all doses B. Geometric mean of the highest non-lethal dose (D_0) and lowest lethal dose (D_100): sqrt(D_0 * D_100) C. Difference between D_100 and D_0 D. Ratio of D_100 to D_0

Answer: B

24. What is the major ethical advantage of Lorke's method over classical probit analysis?

A. It uses far fewer experimental animals (typically 13 mice instead of 50–100 mice) B. It does not require weighing the animals C. It eliminates the need for calculating doses D. Animals never experience any toxic effects

Answer: A

25. What is the formula for the Therapeutic Index (TI) of a drug?

A. ED_50 / LD_50 B. LD_50 / ED_50 C. LD_50 * ED_50 D. (LD_50 - ED_50) / 100

Answer: B

26. A high Therapeutic Index (e.g., TI = 1,000) indicates that a drug:

A. Is extremely toxic and dangerous B. Has a very wide margin of safety between effective and lethal doses C. Is completely ineffective clinically D. Cannot be absorbed orally

Answer: B

27. A low Therapeutic Index (e.g., TI < 2.0) indicates that a drug:

A. Has a narrow therapeutic margin and requires therapeutic drug monitoring (e.g., digoxin, lithium, warfarin) B. Can be purchased over the counter without a prescription C. Has zero potential for toxicity D. Does not bind to plasma proteins

Answer: A

28. What is the definition of the Margin of Safety (Certain Safety Factor)?

A. LD_1 / ED_99 B. ED_99 / LD_1 C. LD_50 / ED_50 D. LD_99 / ED_1

Answer: A

29. Why is the Certain Safety Factor (LD_1 / ED_99) clinically superior to the Therapeutic Index (LD_50 / ED_50)?

A. It takes into account the slopes of the dose-response and toxicity curves at the relevant clinical extremes B. It is much easier to divide by zero C. It uses fewer mathematical calculations D. It only applies to intravenous drugs

Answer: A

30. According to the Hodge and Sterner toxicity scale, a substance with an oral LD_50 of < 1 mg/kg is classified as:

A. Practically non-toxic B. Moderately toxic C. Extremely toxic D. Slightly toxic

Answer: C

31. A chemical substance with an oral LD_50 between 50 and 500 mg/kg is classified as:

A. Extremely toxic B. Moderately toxic C. Practically non-toxic D. Relatively harmless

Answer: B

32. A substance with an oral LD_50 > 15,000 mg/kg (15 g/kg) is classified as:

A. Highly toxic B. Moderately toxic C. Practically non-toxic / relatively harmless D. Extremely toxic

Answer: C

33. Which parameter describes toxicity resulting from repeated daily dosing over a period of 1 to 3 months?

A. Acute toxicity B. Subacute / subchronic toxicity C. Chronic toxicity (> 6 months) D. Idiosyncratic toxicity

Answer: B

34. What route of administration in mice generally produces the lowest LD_50 value (highest apparent toxicity) for a systemic drug?

A. Oral (p.o.) B. Intravenous (i.v.) or Intraperitoneal (i.p.) C. Topical dermal application D. Subcutaneous (s.c.)

Answer: B

35. Why is oral LD_50 usually higher than intravenous or intraperitoneal LD_50?

A. Oral drugs are never absorbed B. Incomplete gastrointestinal absorption and first-pass hepatic metabolism reduce the systemic peak concentration C. The stomach destroys 100% of all chemical compounds D. Intravenous injection dilutes the drug instantly

Answer: B

36. In acute toxicity testing, what does the slope of the probit-log dose line signify?

A. The molecular weight of the substance B. The margin of safety; a steep slope indicates that a small increase in dose causes a large increase in mortality C. The solubility of the drug in water D. The speed of renal excretion

Answer: B

37. If an unknown drug administered at 100 mg/kg kills 0/5 mice, and at 200 mg/kg kills 5/5 mice, what can be deduced about the slope?

A. The curve has a very steep slope, indicating a narrow margin between non-lethal and completely fatal doses B. The drug is completely non-toxic C. The curve is completely flat D. The experiment was invalid

Answer: A

38. In a Karber calculation: Group doses are 40, 80, 160, 320 mg/kg with n=5 mice per group. Mortalities are 0/5, 1/5, 3/5, 5/5. What is LD_100?

A. 40 mg/kg B. 80 mg/kg C. 160 mg/kg D. 320 mg/kg

Answer: D

39. What is the fundamental ethical principle in animal research known as the "3Rs"?

A. Repeat, Record, Report B. Replacement, Reduction, and Refinement C. Resuscitate, Recover, Release D. Restrain, Radiate, Remove

Answer: B

40. The ultimate practical conclusion of Experiment 38 is that:

A. Drug toxicity is purely unpredictable and cannot be quantified B. Determining the LD_50 establishes the acute lethal threshold and safety profile required before therapeutic dosing and further drug development can proceed C. All drugs have the same lethal dose regardless of chemical structure D. Animal mortality bears no relationship to human drug safety

Answer: B